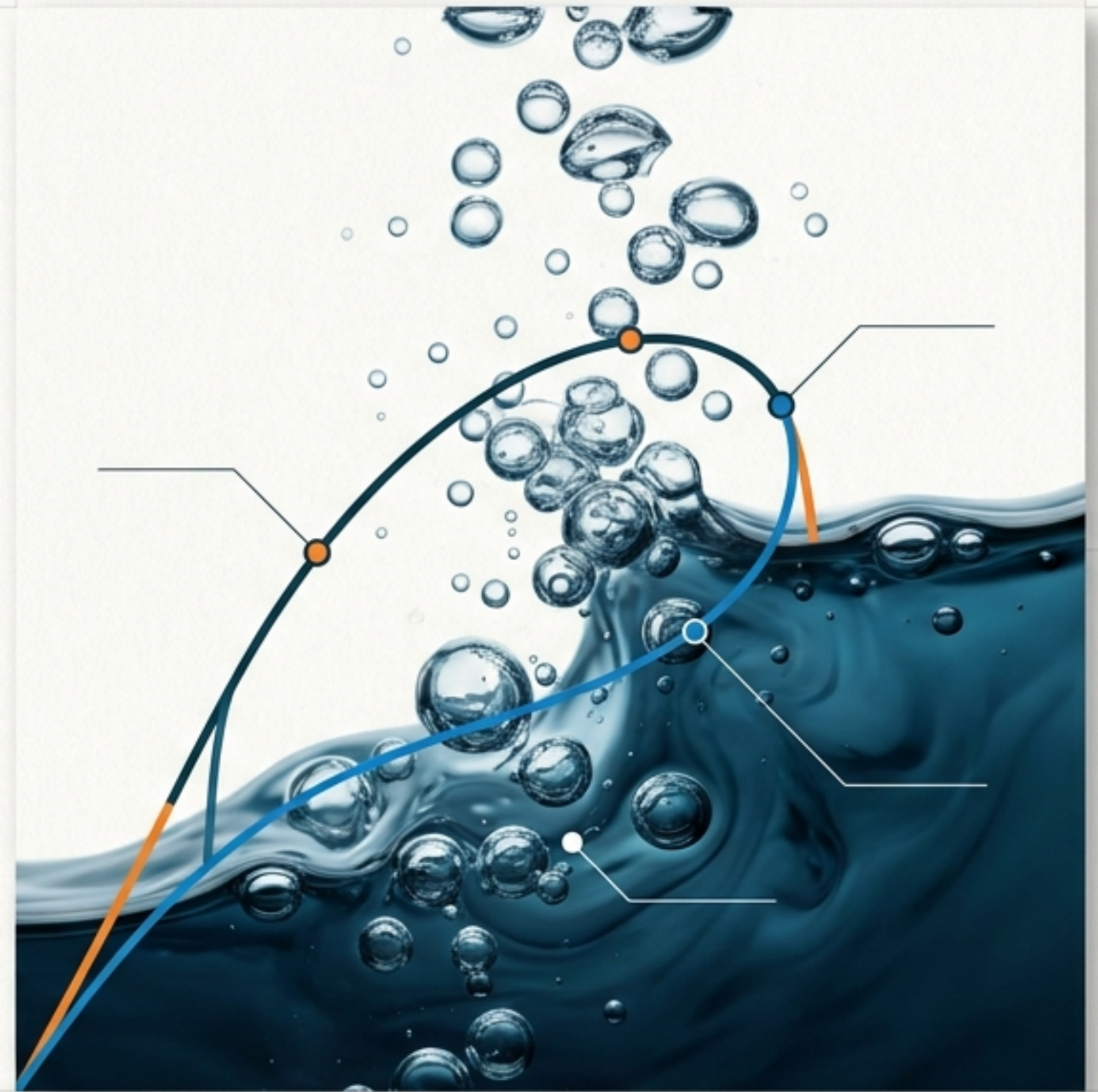


Natural Gas Composition & Phase Behavior

Decoding the PVT Envelope
for Hydrocarbon Recovery

Prepared by
Engineer Sajjad k. Hano



The Illusion of Pure Methane

Consumer Product: Almost 100% Pure Methane (CH₄).



RAW RESERVOIR COMPOSITION

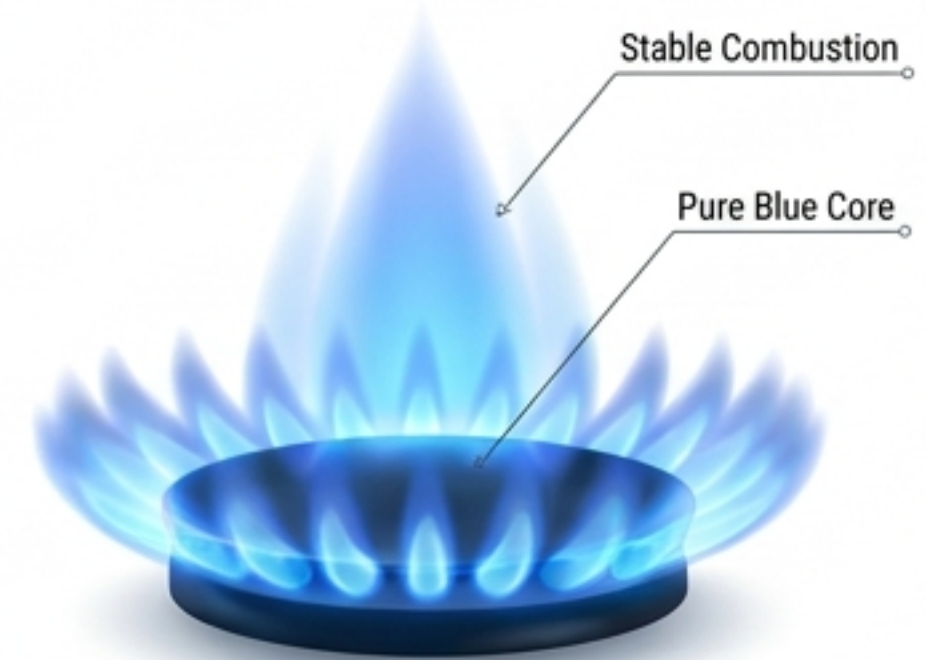
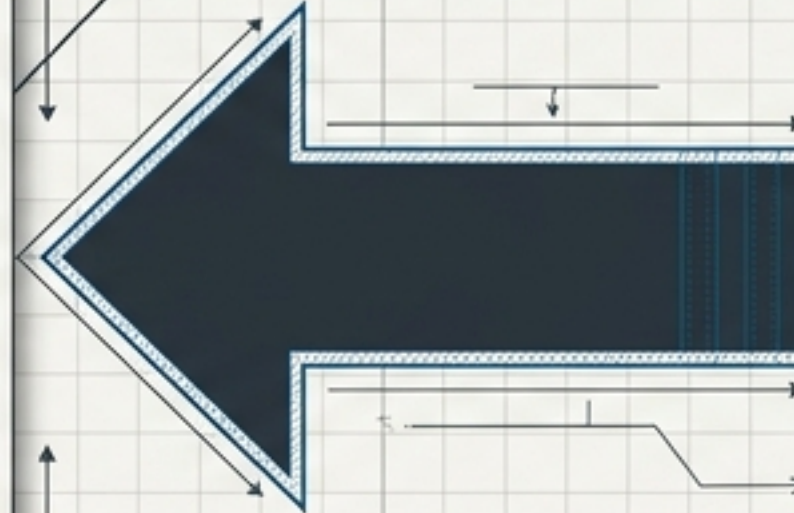
Composition: Highly variable. Requires intense separation.

Primary Hydrocarbons: Methane (CH₄), Ethane (C₂H₆), Propane (C₃H₈), Butane (C₄H₁₀), Pentane (C₅H₁₂).

Nonhydrocarbon Contaminants: Carbon Dioxide (CO₂), Oxygen (O₂), Nitrogen (N₂), Hydrogen Sulphide (H₂S).

Rare Trace Gases: Argon (Ar), Helium (He), Neon (Ne), Xenon (Xe).

Reverse Engineering the Hydrocarbon



CONSUMER PRODUCT SPECIFICATION

Composition: >99.5% Pure Methane (CH₄).

State: Gas.

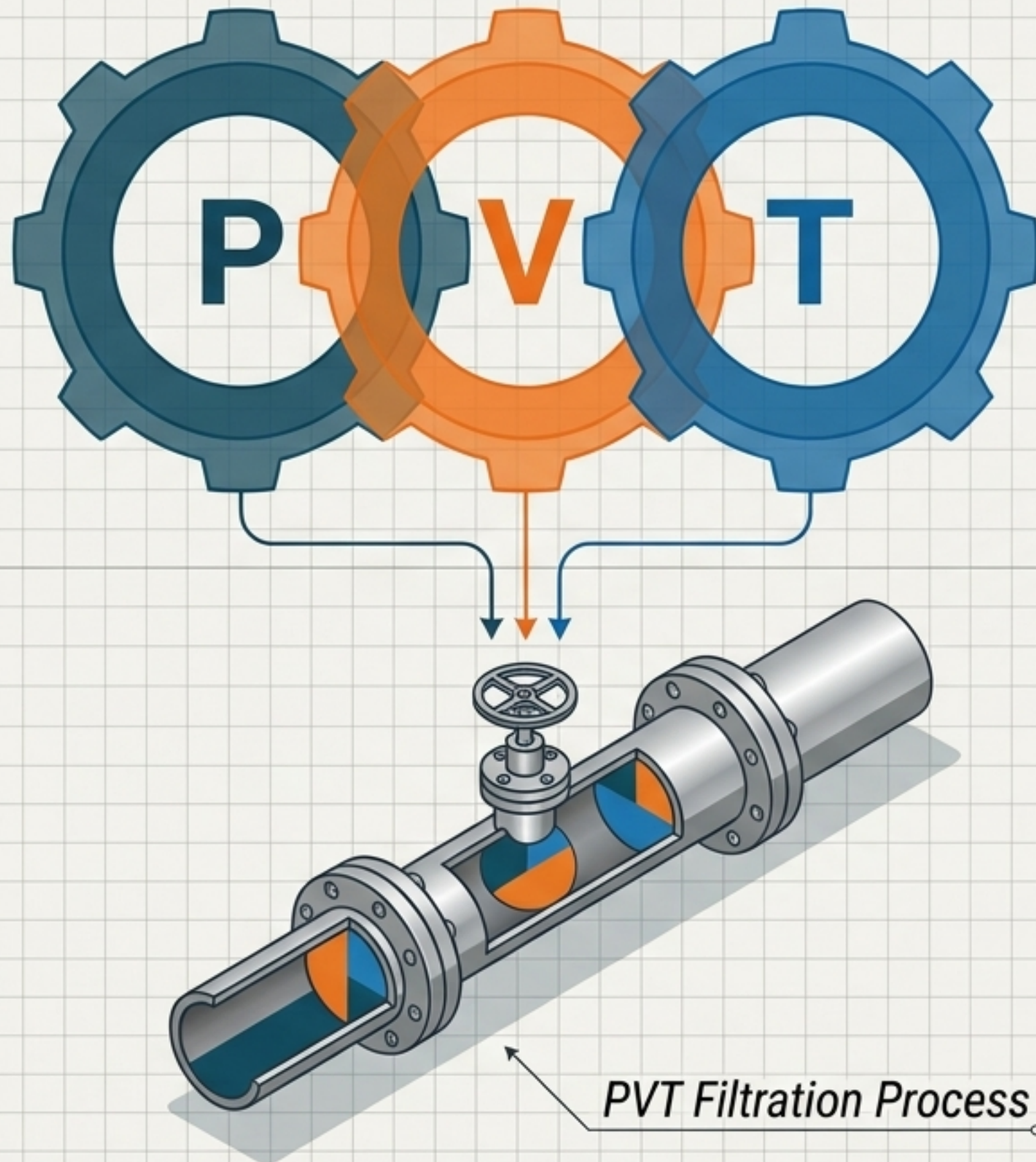
Visual: Clean, stable blue flame.

Application: Domestic and Industrial Heating.

ANALYSIS & PREPARATION

Prepared by Engineer Sajjad k. Hano





The PVT Engine: Controlling State Changes

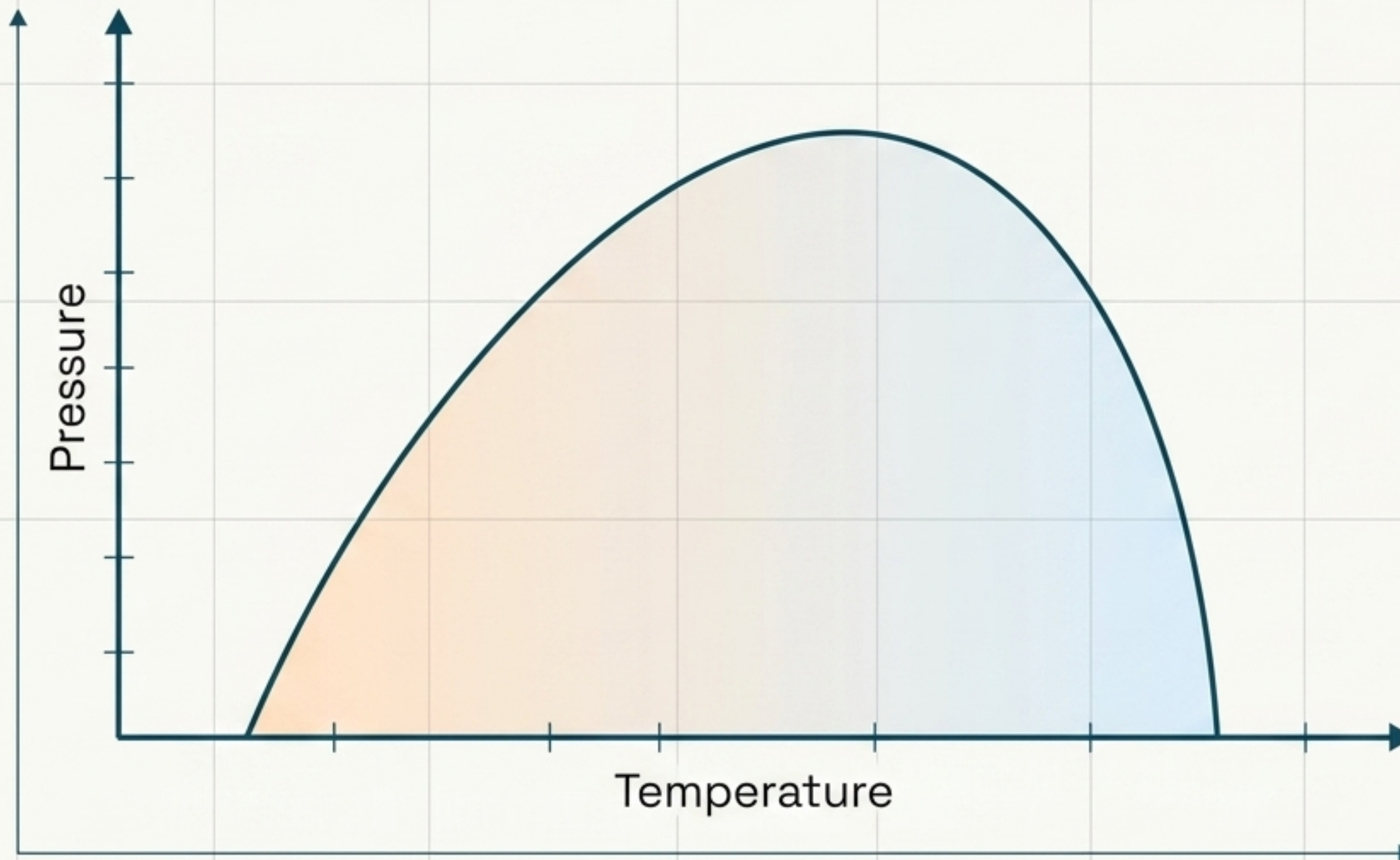
Natural gas phase behavior is strictly a function of **Pressure**, **Temperature**, and **Volume** (PVT).

Understanding the PVT envelope is absolutely critical for:

1. Determining the hydrocarbon recovery mechanism.
2. Executing accurate production prediction.

The Phase Canvas: The Two-Phase Region

Two-Phase Region



Key Takeaway:

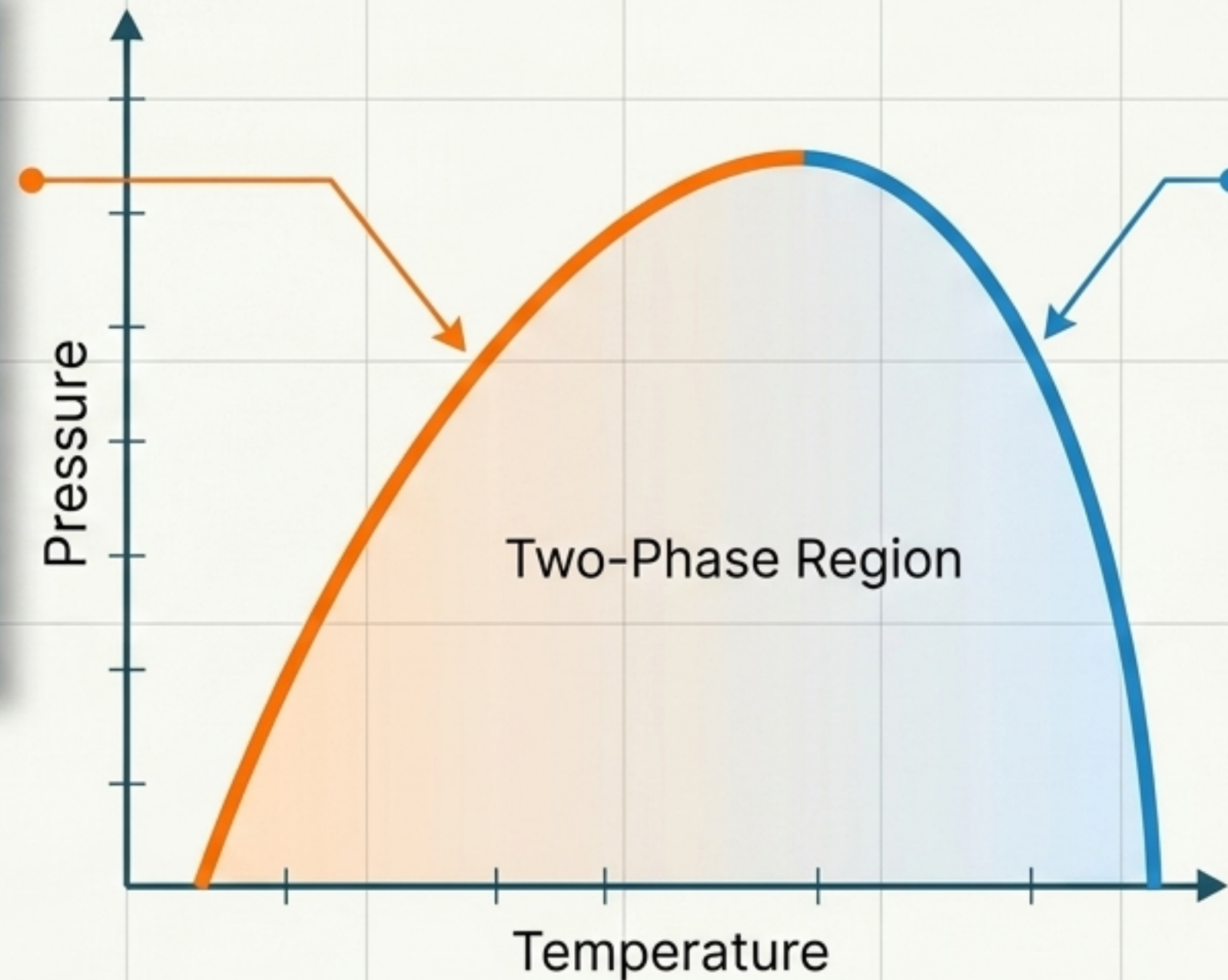
This central envelope dictates where **multiple states** of matter exist **simultaneously**. Understanding the boundaries of this shape is the key to reservoir management.

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Borderlines of State Change

Bubble Point Curve

- Separates pure **liquid (oil)** from the two-phase region.
- The **100% Liquid Boundary**.
- **Mechanism:** At a given temperature, as pressure decreases below this curve, gas is emitted from the liquid phase.



Dew Point Curve

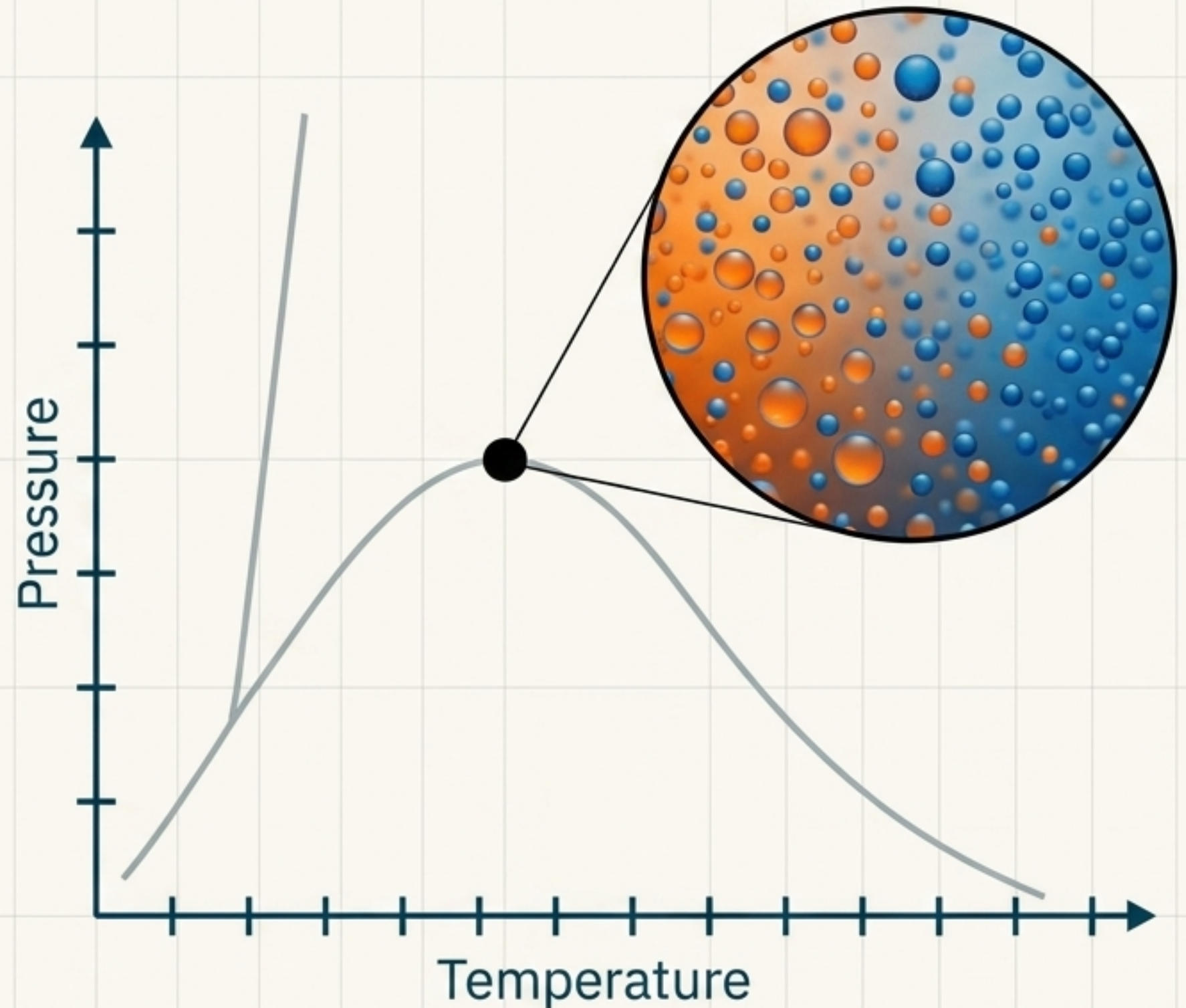
- Separates pure **gas** from the two-phase region.
- The **0% Liquid Boundary**.
- **Mechanism:** The connected points of P and T where the absolute first liquid droplet forms out of the gas phase.

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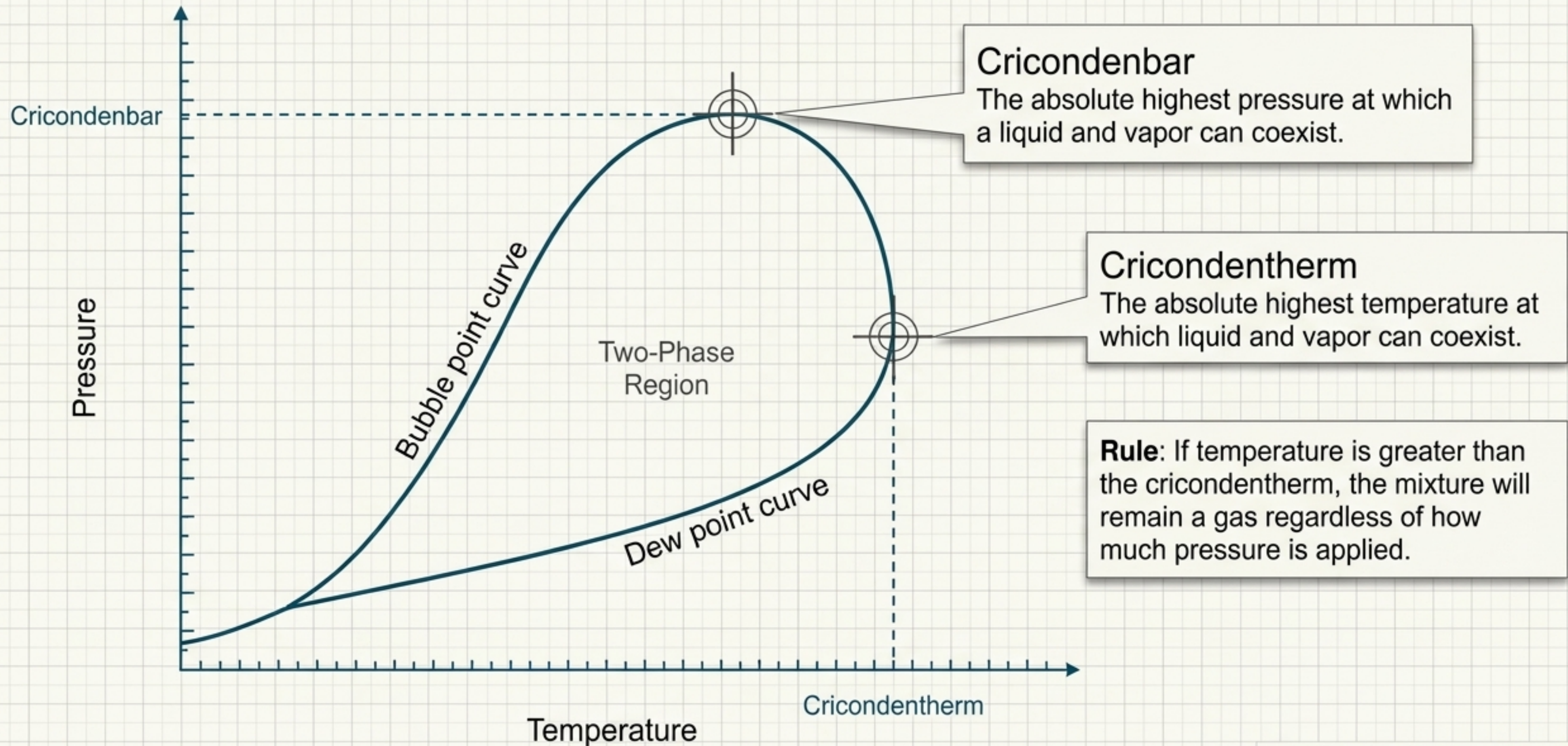
The Apex of Convergence: The Critical Point

Definition: The exact coordinate on the phase envelope where the Bubble Point curve meets the Dew Point curve.

The Physics Break: At this specific critical pressure (P_c) and critical temperature (T_c), the physical properties of the gas become absolutely identical to the properties of the liquid.



The Outer Limits of Coexistence



Synthesizing the PVT Blueprint

Parameter	Classification	State Boundary / Transition	Critical Limit
Bubble Point	Curve	Pure Liquid -> Two-Phase (Gas emits)	100% Liquid Boundary
Dew Point	Curve	Pure Gas -> Two-Phase (First drop forms)	0% Liquid Boundary
Critical Point	Coordinate (P _c , T _c)	Gas Properties = Liquid Properties	Point of Convergence
Cricondenbar	Maximum Limit	Liquid/Vapor Coexistence	Absolute Highest Pressure
Cricondentherm	Maximum Limit	Liquid/Vapor Coexistence	Absolute Highest Temperature

